	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	1715 - softmax_v6_vec4	(8192, 1, 1)x(256, 1, 1)	660.06 us	17,33,186	0 - NVIDIA GeForce RTX 5080	2.62 Ghz	[937025] softmax	

Summary | Details | Source | Context | Comments | Raw | Session

Compare | Tools | View | Export

GPU Speed Of Light Throughput

All

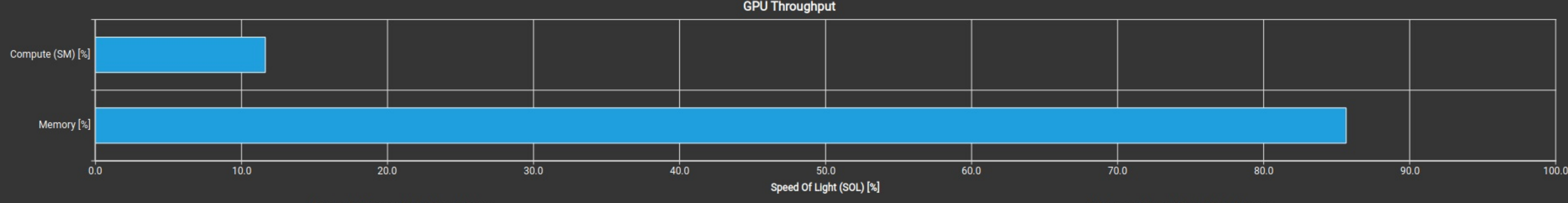
High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	11.64	Duration [us]	660.06
Memory Throughput [%]	85.65	Elapsed Cycles [cycle]	17,33,186
L1/TEX Cache Throughput [%]	14.47	SM Active Cycles [cycle]	16,54,402.96
L2 Cache Throughput [%]	25.27	SM Frequency [Ghz]	2.62
DRAM Throughput [%]	85.65	DRAM Frequency [Ghz]	14.79

High Throughput This workload is utilizing greater than 80.0% of the available compute or memory performance of this device. To further improve performance, work will likely need to be shifted from the most utilized to another unit. Start by analyzing DRAM in the Memory Workload Analysis section.

Roofline Analysis The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved 2% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Profiling Guide for more details on roofline analysis.

GPU Throughput



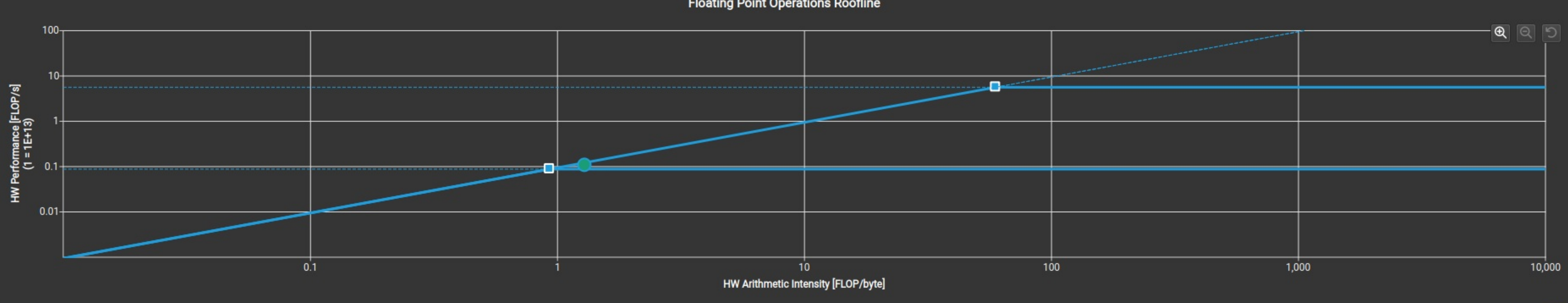
Compute Throughput Breakdown

SM: Inst Executed [%]	11.64
SM: Issue Active [%]	11.64
SM: Inst Executed Pipe Xu [%]	9.81
SM: Pipe Fmaheavy Cycles Active [%]	7.39
SM: Pipe Fma Cycles Active [%]	6.07
SM: Inst Executed Pipe Lsu [%]	3.80
SM: Pipe Aluheavy Cycles Active [%]	3.59
SM: Mio Pq Write Cycles Active [%]	2.01
SM: Mio Pq Read Cycles Active [%]	2.00
SM: Mio Inst Issued [%]	1.63
SM: Inst Executed Pipe Adu [%]	1.10
SM: Mio2rf Writeback Active [%]	0.94
SM: Inst Executed Pipe Cbu Pred On Any [%]	0.55
IDC: Request Cycles Active [%]	0.36
SM: Memory Throughput Internal Activity [%]	0.27
SM: Inst Executed Pipe Uniform [%]	0.19
SM: Instruction Throughput Internal Activity [%]	0
SM: Inst Executed Pipe Tex [%]	0
SM: Inst Executed Pipe Ipa [%]	0
SM: Pipe Fp64 Cycles Active [%]	0
SM: Pipe Tensor Cycles Active [%]	0
SM: Pipe Tma Cycles Active [%]	0

Memory Throughput Breakdown

DRAM: Cycles Active [%]	85.65
DRAM: Dram Sectors [%]	67.24
L2: T Sectors [%]	25.27
L2: Xbar2lts Cycles Active [%]	21.19
L2: D Sectors [%]	20.91
L2: D Sectors Fill Device [%]	17.25
L2: Lts2xbar Cycles Active [%]	16.83
GPU: Compute Memory Access Throughput Internal Activity [%]	14.99
L2: T Tag Requests [%]	12.66
L1: M Xbar2l1tex Read Sectors [%]	11.52
L1: M L1tex2xbar Req Cycles Active [%]	7.23
L1: Data Pipe Lsu Wavefronts [%]	5.41
L1: Lsuin Requests [%]	3.80
L1: Lsu Writeback Active [%]	3.56
L1: Data Bank Reads [%]	2.17
L1: Data Bank Writes [%]	2.16
SYSLTS: T Sectors [%]	0.02
SYSLTS: Xbar2syslts Cycles Active [%]	0.02
GPU: Compute Memory Request Throughput Internal Activity [%]	0.02
SYSLTS: T Tag Requests [%]	0.01
SYSLTS: D Sectors [%]	0.01
L1: Texin Sm2tex Req Cycles Active [%]	0.00
SYSLTS: Syslts2xbar Cycles Active [%]	0.00
SYSLTS: D Sectors Fill Sysmem [%]	0.00
L2: D Decomp Input Sectors [%]	0
L2: D Atomic Input Cycles Active [%]	0
L1: Tmain Requests [%]	0
L1: Tex Writeback Active [%]	0
L1: M Xbar2l1tex Read Sectors Mem Dshared [%]	0
L1: F Wavefronts [%]	0
SYSLTS: D Atomic Input Cycles Active [%]	0
L1: Data Pipe Tex Wavefronts [%]	0

Floating Point Operations Roofline



Compute Workload Analysis

Pipe Utilization (Elapsed Cycles)

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

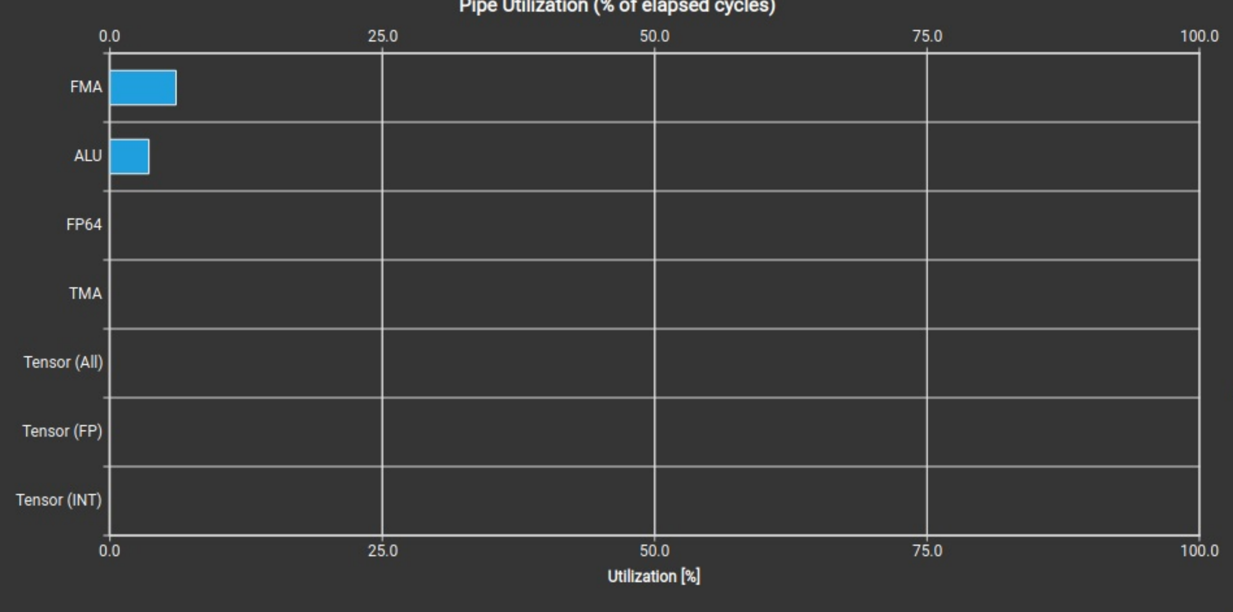
Executed Ipc Elapsed [inst/cycle]	0.47	SM Busy [%]	11.64
Executed Ipc Active [inst/cycle]	0.49	Issue Slots Busy [%]	11.64
Issued Ipc Active [inst/cycle]	0.49		

Low Utilization
Est. Local Speedup: 93.93%

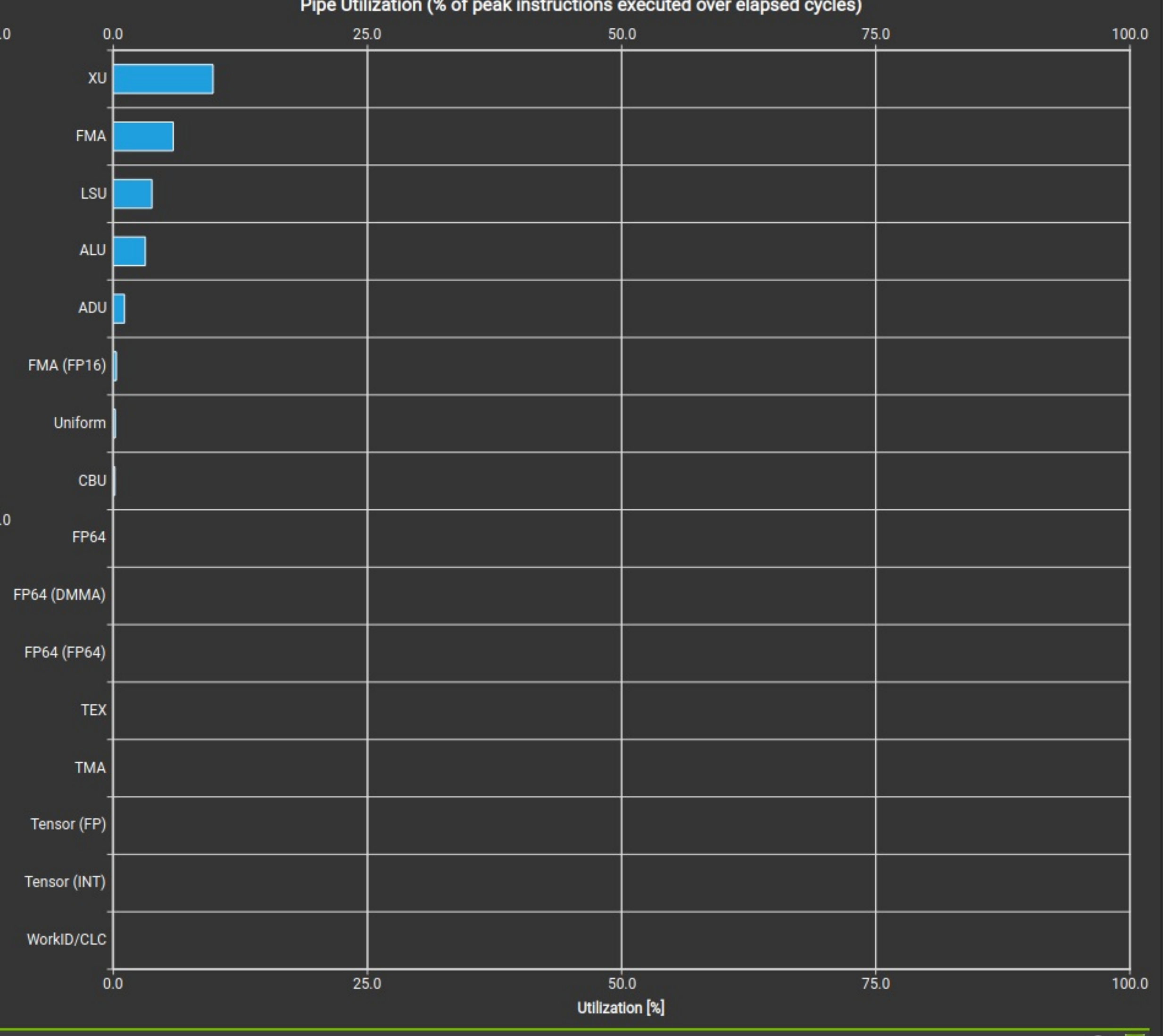
All compute pipelines are under-utilized. Either this workload is very small or it doesn't issue enough warps per scheduler. Check the Launch Statistics and Scheduler Statistics sections for further details.

Key Performance Indicators

Pipe Utilization (% of elapsed cycles)



Pipe Utilization (% of peak instructions executed over elapsed cycles)



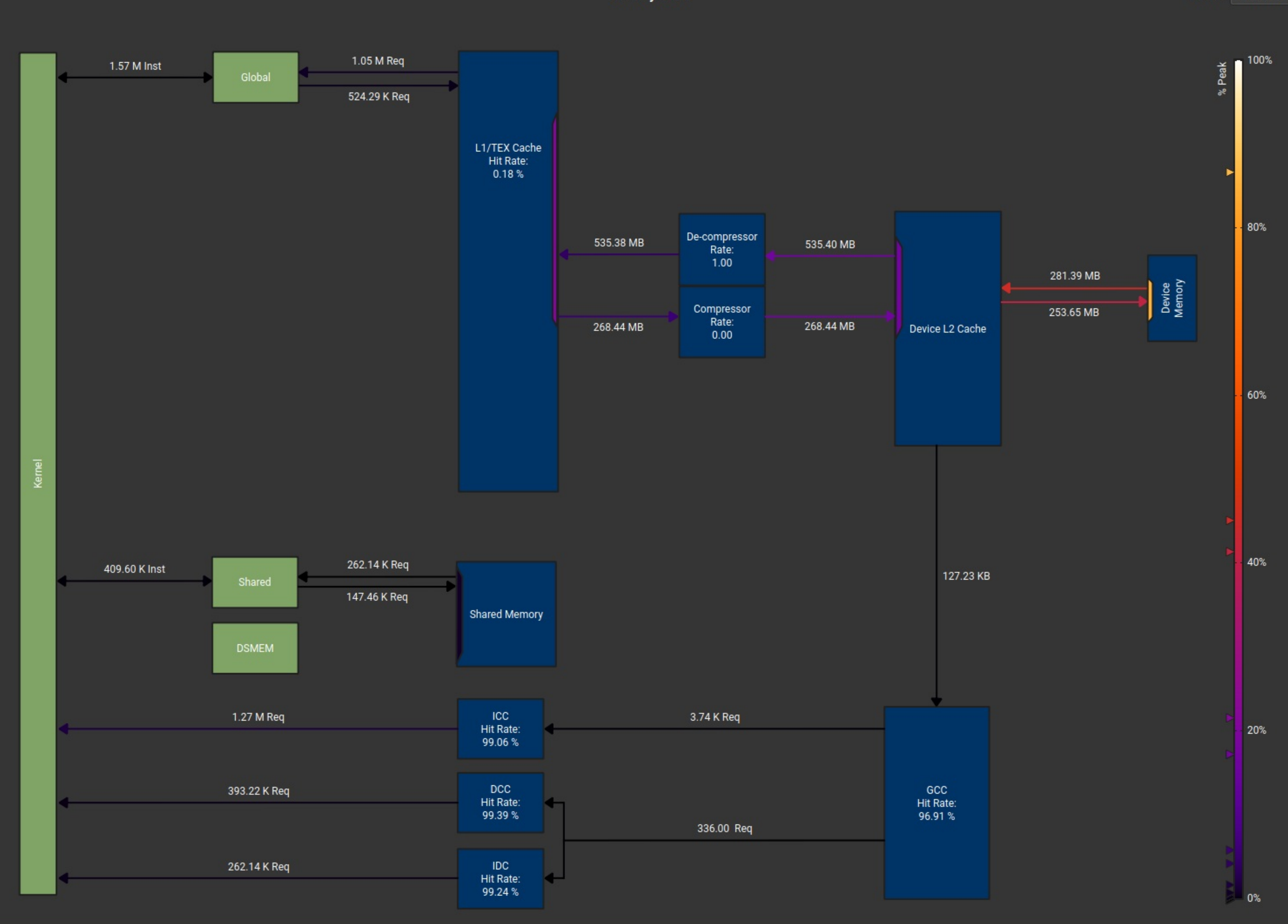
Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units.

Memory Throughput [Gbyte/s]	810.59	Mem Busy [%]	25.27
L1/TEX Hit Rate [%]	0.18	Max Bandwidth [%]	85.65
L2 Hit Rate [%]	32.48	Mem Pipes Busy [%]	3.80
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Shared Memory Spilling Requests	0
L2 Compression Success Rate [%]	0	Local Memory Spilling Request Overhead [%]	0
L2 Persisting Size [Mbyte]	12.58	Shared Memory Spilling Request Overhead [%]	0
L2 Sector Promotion Misses [%]	0	-	-

Memory Chart

Values: Transfer Size Inactivity: Hidden



Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	8,192	Function Cache Configuration	CachePreferNone
Cluster Size	0	Preferred Cluster Size	0
Registers Per Thread [register/thread]	37	Cluster Scheduling Policy	PolicySpread
Static Shared Memory Per Block [byte/block]	72	Block Size	256
Dynamic Shared Memory Per Block [byte/block]	0	Threads [thread]	20,97,152
Driver Shared Memory Per Block [Kbyte/block]	1.02	Waves Per SM	16.25
Shared Memory Configuration Size [Kbyte]	16.38	Uses Green Context	0
Stack Size	1,024	# SMs [SM]	84
# TPCs	42	Enabled TPC IDs	all

WorkID/CLC Info

	Average	Min	Max	Sum
WorkId/CLC Requests	0	0	0	0
WorkId/CLC Requests Granted	0	0	0	0
WorkId/CLC Requests Granted as CTAs	0	0	0	0

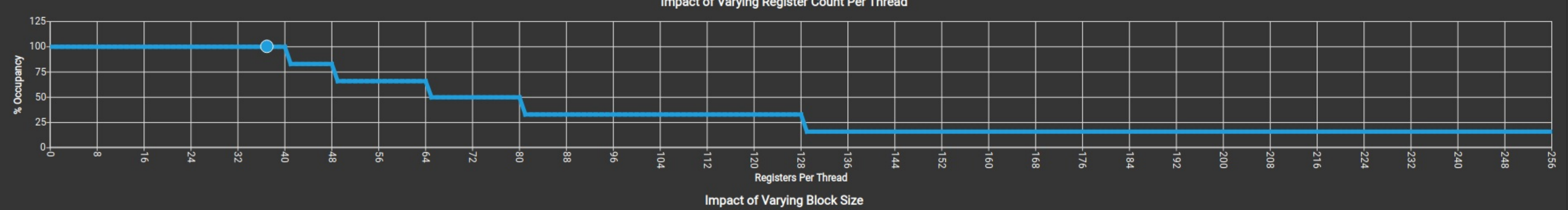
Occupancy

% Occupancy Graphs

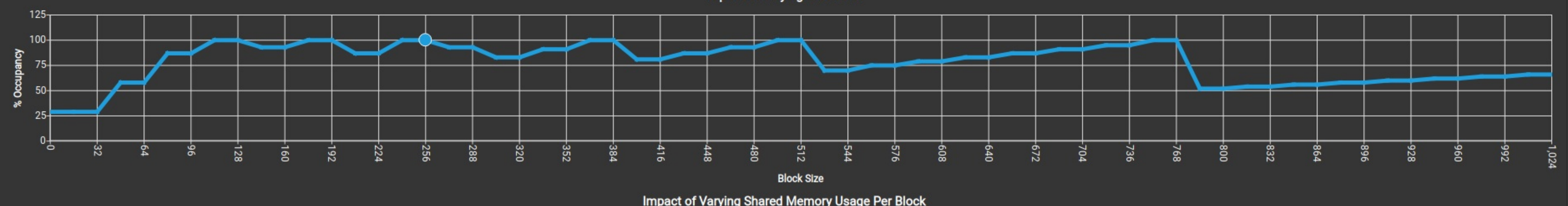
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	6
Theoretical Active Warps per SM [warp]	48	Block Limit Shared Mem [block]	14
Achieved Occupancy [%]	95.45	Block Limit Warps [block]	6
Achieved Active Warps Per SM [warp]	45.82	Block Limit SM [block]	24
Cluster Occupancy [%]	0	Block Limit Barriers [block]	24
Max Active Clusters [cluster]	0	Max Cluster Size [block]	8
Overall GPU Occupancy [%]	0		

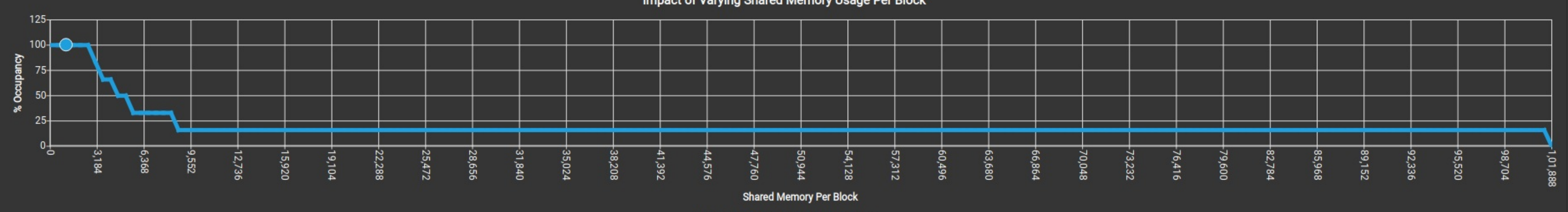
Impact of Varying Register Count Per Thread



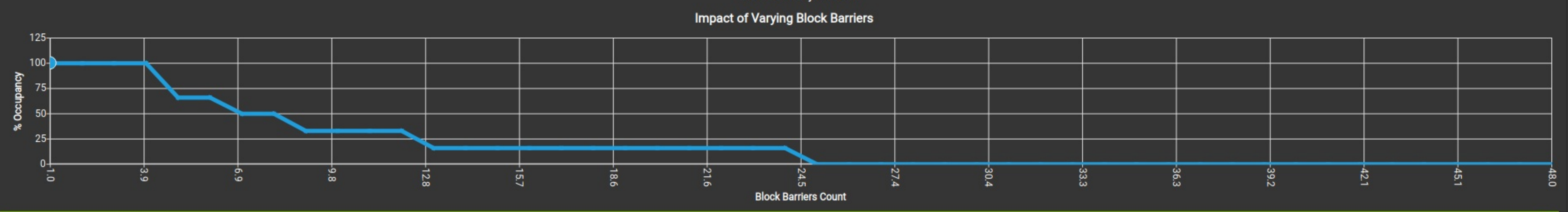
Impact of Varying Block Size



Impact of Varying Shared Memory Usage Per Block



Impact of Varying Block Barriers



GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	16,54,402.96	Average L1 Active Cycles [cycle]	16,54,402.96
Average L2 Active Cycles [cycle]	14,46,228.66	Average SMSP Active Cycles [cycle]	16,58,173.93
Average DRAM Active Cycles [cycle]	83,60,032	Total SM Elapsed Cycles [cycle]	14,53,05,816
Total L1 Elapsed Cycles [cycle]	14,53,05,816	Total L2 Elapsed Cycles [cycle]	4,97,29,120
Total SMSP Elapsed Cycles [cycle]	58,12,23,264	Total DRAM Elapsed Cycles [cycle]	7,80,81,024

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	16,54,402.96	16,26,005	16,61,949	13,89,69,849
SMSP Active Cycles	16,58,173.93	16,38,007	16,65,999	55,71,46,442
L1 Active Cycles	16,54,402.96	16,26,005	16,61,949	13,89,69,849
L2 Active Cycles	14,46,228.66	14,44,012	14,46,939	4,62,79,317
DRAM Active Cycles	83,60,032	82,96,576	84,20,992	6,68,80,256

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	20,31,616	Branch Efficiency [%]	95.08
Branch Instructions Ratio [%]	0.03	Avg. Divergent Branches [branches]	219.43

Warp Stall Sampling (Not-issued Samples)

Location	Value	Value (%)
softmax.cu:291 (0x70bfe376b820 in softmax_v6_v...	30,807	51
softmax.cu:325 (0x70bfe376e220 in softmax_v6_v...	22,205	37
softmax.cu:277 (0x70bfe376cc00 in softmax_v6_v...	4,537	8
softmax.cu:320 (0x70bfe376e000 in softmax_v6_v...	403	1
softmax.cu:322 (0x70bfe376e1c0 in softmax_v6_v...	374	1

Most Instructions Executed

Location	Value	Value (%)
softmax.cu:321 (0x70bfe376e410 in softmax_v6_v...	5,24,288	1
softmax.cu:328 (0x70bfe376e400 in softmax_v6_v...	5,24,288	1
softmax.cu:327 (0x70bfe376e3f0 in softmax_v6_ve...	5,24,288	1
softmax.cu:327 (0x70bfe376e3e0 in softmax_v6_v...	5,24,288	1
softmax.cu:326 (0x70bfe376e3d0 in softmax_v6_v...	5,24,288	1

Still missing data? Collect the *full* metric set, enable individual sections or learn how to select individual metrics for collection.

You can ask Nsight Copilot questions about this report.